

3/18 Group Meeting 中橋 - 萌 Hayime Nakahashi
The See-Saw Principle and its Applications to Abelian Varieties.

ref. D. Mumford "Abelian Varieties"

Ch I, Ch II, 9 [A] Ch II [V] Ch III [S]

• S. Lang "Abelian Varieties" [V]

• S. Birkenhake & H. Lange "Complex Abelian varieties" [A]

• A. Polishchuck "Abelian varieties ..." Ch I [A] Ch II [V]

* Abelian var. (and line bundles on it) can be considered from various perspectives.

[A]: Analytic $\rightarrow$ using Appel-Humbert datum.
(H, α) H : Hermitian form α : semi-char.

[V]: via variety $\rightarrow$ today

[S]: via scheme $\rightarrow$ for situation for char = p.

Motivation

To Investigate "Vector bundles on abelian var."

- What kind of vect. bundle exist?
- How are they classified?

$\rightarrow$ Today (and next time ...)'s Goal is

to construct some canonical line bundle on $A \times \hat{A}$ (Poincaré bundle) and to explain what is dual of Abelian var.

Def of Abelian VarMotivationcpt, conn. cpx Lie gr $\cong$ cpx toriAnalytic: X : proj alg var is Abelian if $X \cong V/T$: cpx tori as cpx mtd.where V : n -dim $\mathbb{C}$ -vect sp. $T \subset \mathbb{C}^n$: free Abelian gr rank $2n$.Q Every cpx torus is abelian var? $\rightarrow$ No! it can be checked by Riemann-Relations $\rightarrow$ See Birk & Lange Ch 4.Via Var: complete alg var with group law. $m: X \times X \rightarrow X$ such that m and the inverse map are both morphism of varieties.

How often/naturally abelian var. appears?

(sm complete)

var V Picard var: $\text{Pic}^0(V)$ Albanese var: $\text{Alb}(V)$

} Abelian var!

① $\text{Pic}^0(V)$: connected comp of identity in Picard sch. $\cong \text{ker } C_1: H^1(\mathcal{O}^*) \rightarrow H^2(\mathbb{Z})$

topologically trivial line bundle.

e.g curve: Jacobian Var.

surf: (char=0) dim = irregularity of a surf.

② $\text{Alb}(V)$: dual of hol. 1-forms modulo period
characterized by univ. prop $V \rightarrow \text{Alb}(V)$ $\downarrow$ morph $\searrow$ $\mathbb{Z}$ $\downarrow$ Ab. Var

e.g surf: (char=0)

dim = irregularity of surf.

Relations between $\text{Pic}^0(V)$ & $\text{Alb}(V)$

For curve: Abel-Jacobi Thm
 $\text{Alb}(C) \simeq \text{Pic}^0(C)$

For higher dim:

$$\text{Alb}(V) \simeq \widehat{\text{Pic}^0(V)} \leftarrow \text{dual of an Abelian var.}$$

What is dual?

X : ab. var. $\rightarrow$ we define $\hat{X} = \text{Pic}^0(X)$

Moreover we can write more explicitly

Analytic: $\hat{X} \cong \mathbb{C}^n / \Gamma^n$ where $\Gamma^n = \{f \in \mathbb{C}^n \mid 2\text{Re}(f) \in \mathbb{Z}\}$

$\hookrightarrow$ this is cpx tori (See Griffiths, Harris Ch6)

Via Var.: $\hat{X} \simeq X / \ker \phi_L$ ϕ_L : isogeny.

$\hookrightarrow$ this is Abelian var. (later...)

What property would we expect, which would characterize this "extra structure"?

- (a) We want line bundle P on $X \times \hat{X}$
 s.t. $\begin{cases} \forall \alpha \in \hat{X} & P|_{X \times \{\alpha\}} = \alpha \in \text{Pic}^0(X) \quad (\text{relevant cond.}) \\ \text{for } 0 \in X & P|_{\{0\} \times \hat{X}} = \text{triv.} \quad (\text{normalization.}) \end{cases}$

such line bundle is called Poincaré line bundle.

- (b) (univ. prop) For every normal var S
 every line bundle K on $X \times S$
 s.t. $\begin{cases} \bullet K|_{X \times \{s\}} \text{ is in } \text{Pic}^0(X) \text{ for } \forall s \in S \\ \bullet K|_{\{0\} \times \hat{X}} \text{ is triv} \end{cases}$

Then $\exists f: S \rightarrow \hat{X}$ set theoretic

s.t. $K_s \simeq P(f(s))$, f is morph, $K \simeq (1_X \times f)^* P$

$\rightarrow$ This implies P is the solution of the moduli problem. CHA SHIN

Note

Poincaré line bundle appears as the F-M kernel of equiv.

$$\mathbb{P}: \mathcal{D}^b(\hat{X}) \longrightarrow \mathcal{D}^b(X) \quad X: \text{abel. var.}$$

by Mukai.

To construct & to check uniqueness of P , we need the See-Saw Principle & its applications.

Idea: We can relate properties of line bundles on $X \times Y$ to their restrictions to $X \times \{y\}$ & $\{x\} \times Y$.
 Looking bundles from two angles

Prop X : irred, complete var.

T : integral sch.

$$X_t \doteq X \times \{t\}.$$

$L \in \text{Pic}(X \times T)$ suppose $L_t \doteq L|_{X_t}$ is trivial.

Then there exists a line bundle M on T .

with $L \simeq p^*M$ where $p: X \times T \longrightarrow T$.

pf By assumption $h^0(L_t) \equiv 1$ const. X_t : complete.

then by Grauert (semicontinuity & base change)
 see Hartshorne Ch II.12

$$H^0(X_t, L_t) \simeq p_* L \otimes_{\mathcal{O}_T} k(t)$$

is (by $L_t \simeq \mathcal{O}_{X_t}$)

$$k(t) \implies p_* L: \text{line bundle. (Let } M \doteq p_* L)$$

Now we have the adjunction map

$$\alpha: p^* p_* (L) \longrightarrow L$$

claim α : surjective

Take fibers

$$\begin{array}{ccc}
 \alpha_t: P^* P_* L|_t \longrightarrow L_t & & \\
 \downarrow \text{is} & \searrow \text{is} & \\
 \mathcal{O}_{X_t} \otimes_{k(t)} T(X_t, L_t) & \xrightarrow{\quad} & \mathcal{O}_{X_t} \\
 \downarrow \text{is} & \Rightarrow \alpha|_t \text{ isom.} & \\
 \mathcal{O}_{X_t} & &
 \end{array}$$

Note	stalk.
$\mathcal{F} _t$	$\mathcal{F}_t / m^+ \mathcal{F}_t$
tiber	

$$\begin{aligned}
 &\Rightarrow \alpha|_{(x,t)}: P^* P_* L|_{(x,t)} \longrightarrow L|_{(x,t)} \text{ is also surj in } (\mathcal{O}_{X,T,(x,t)}, \mathcal{M}) \\
 &\text{NAK} \quad \text{stalk} \quad \text{local ring} \\
 &\Rightarrow \alpha_{(x,t)}: P^* P_* L|_{(x,t)} \longrightarrow L|_{(x,t)} \text{ is surj} \\
 &\Rightarrow \alpha: P^* P_* L \longrightarrow L \text{ is surj.}
 \end{aligned}$$

Since $L \in P^* P_* L$ is loc. free, and compare ranks we get α is isom $\star$

Cor1 L, L' line bundle on $X \times T$
 Suppose $L_t \cong L'_t$ for all $t \in T$
 Then $L \cong L' \otimes P^* M$ for some $M \in \text{Pic}(T)$

pf apply P_{wp} to $L \otimes L'^{-1}$ $\star$

Cor2 (See-Saw Principle)

Suppose that in addition to Cor1 $L_x \cong L'_x$ for at least one $x \in X$, Then $L \cong L'$

pf from Cor1, $L \cong L' \otimes P^* M$

we pullback by $t \mapsto (x, t)$, we get

$$L|_x \cong L'|_x \otimes (P^* M)|_x$$

isom

$\Delta(P^* M)_x = M$ Then M is triv. $\star$

Application (Theorem of the cube)

$X \times Y \times Z$: product of three irred. complete var.
with chosen $x_0 \in X, y_0 \in Y, z_0 \in Z$.

Then line bundle L on $X \times Y \times Z$ is triv

$\Leftrightarrow$ the three restrictions

$L|_{\{x_0\} \times Y \times Z}, L|_{X \times \{y_0\} \times Z}, L|_{X \times Y \times \{z_0\}}$: triv.

pf Next time.

(I'll refer Polishchuck. Thm 8.3)